

TABLE I
Supplementary Table S1. Full per-condition statistics for matched QNN-control comparisons.

Regime	Control F1	macro- No-ent statistics	ZZ+RA statistics	Verdict
TL4 / small CNN / 16- shot	0.3055 ± 0.1729	0.5031 ± 0.1421 ; $\Delta = +0.1976$; $p = 0.03125$; CI $[+0.0766, +0.3271]$; signs 6/0/1; $\rho = 2.30\times$	0.4545 ± 0.1735 ; $\Delta = +0.1491$; $p = 0.01562$; CI $[+0.0621, +0.2665]$; signs 7/0/0; $\rho = 5.51\times$	Inside QCAD
TL4 / small CNN / 32- shot	0.6034 ± 0.0478	0.6128 ± 0.0557 ; $\Delta = +0.0094$; $p = 0.81250$; CI $[-0.0081, +0.0302]$; signs 3/0/4; $\rho = 2.24\times$	0.6053 ± 0.0239 ; $\Delta = +0.0018$; $p = 0.81250$; CI $[-0.0265, +0.0257]$; signs 4/0/3; $\rho = 6.30\times$	Gain contracts
TL4 / small CNN / 64- shot	0.6705 ± 0.0378	0.6685 ± 0.0280 ; $\Delta = -0.0021$; $p = 0.81250$; CI $[-0.0144, +0.0068]$; signs 4/0/3; $\rho = 2.44\times$	0.6707 ± 0.0314 ; $\Delta = +0.0001$; $p = 0.93750$; CI $[-0.0045, +0.0058]$; signs 3/0/4; $\rho = 7.47\times$	Gain vanishes
TL4 / ResNet18 / 16- shot	0.8545 ± 0.0059	0.8534 ± 0.0027 ; $\Delta = -0.0011$; $p = 0.68750$; CI $[-0.0049, +0.0034]$; signs 2/0/5; $\rho = 1.58\times$	0.8596 ± 0.0125 ; $\Delta = +0.0051$; $p = 0.57812$; CI $[-0.0013, +0.0133]$; signs 3/0/4; $\rho = 3.90\times$	Compressed
Urban4 / small CNN / 16-shot	0.4655 ± 0.1655	0.4593 ± 0.1951 ; $\Delta = -0.0063$; $p = 0.68750$; CI $[-0.1599, +0.1398]$; signs 2/0/5; $\rho = 2.17\times$	0.4073 ± 0.1658 ; $\Delta = -0.0583$; $p = 0.15625$; CI $[-0.1264, -0.0087]$; signs 2/0/5; $\rho = 5.32\times$	No stable trans-fer

Supplementary Material for “Regime-Bounded Metrics for Qiskit Hybrid Quantum Heads in Low-Shot Remote Sensing ROI Classification”